Senior High School

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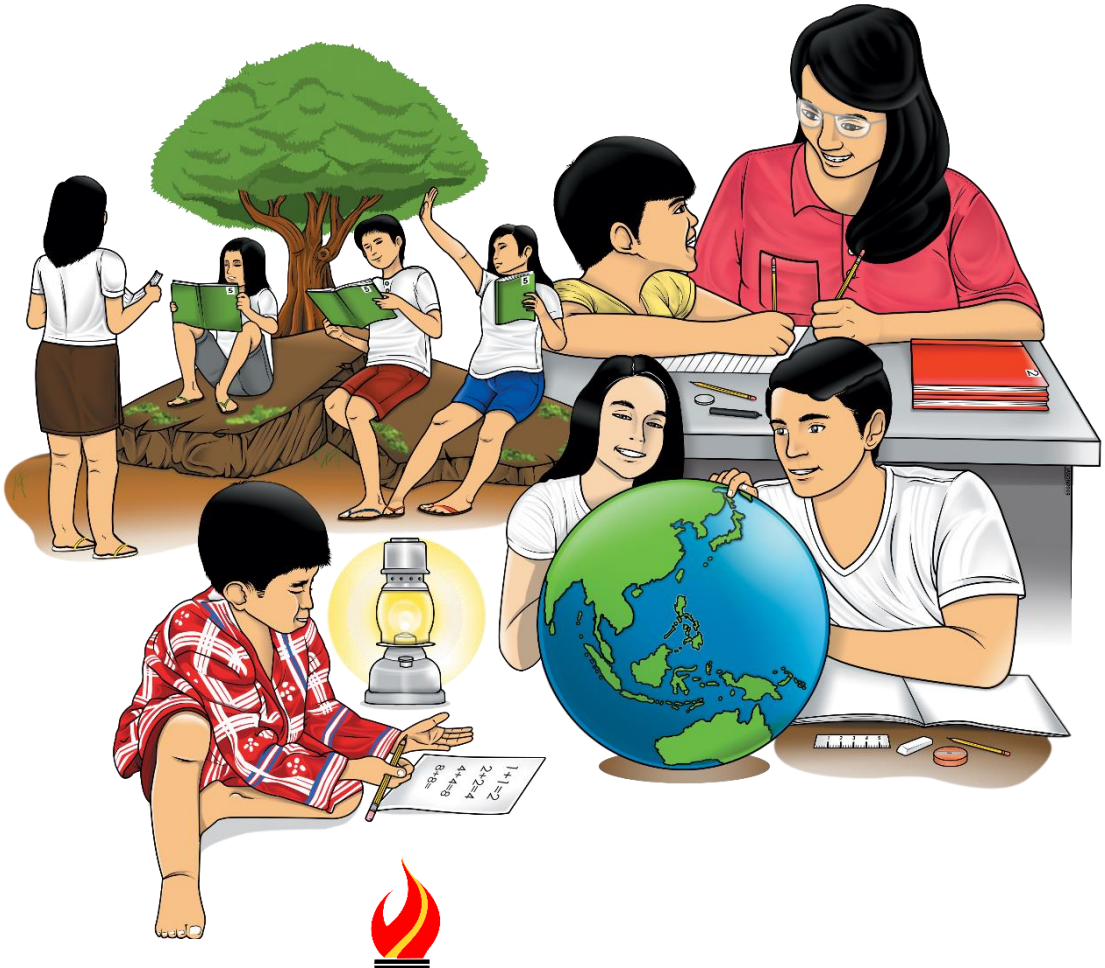
PRE-CALCULUS
Quarter 2



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Pre- Calculus
Alternative Delivery Mode
Quarter 2 – Module
First Edition, 2020

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What I Need to Know

Expected Learning Outcomes: **(LC Codes: STEM_PC11AG-IIa-1-2)**

1. Illustrate the unit circle and the relationship between the linear and angular; and
2. Convert degree measure to radian measure and vice versa.

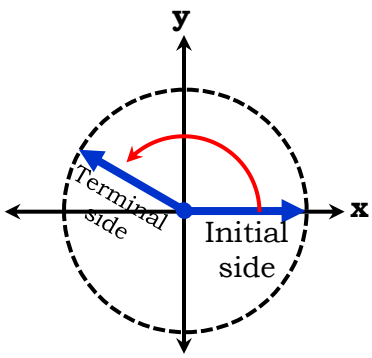
What I know

Choose the letter of the correct answer.

1. **A point on the circumference of the unit circle has coordinates $\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$.**
   What angle does this represent in degree?
   A. 30º B. 45º C. 60º D. 90º
2. **Convert 120° to radians.**
   A. 2.10 rad B. 2.03 rad C. 2.9 rad D. 2.09 rad
3. **$\frac{3\pi}{4}$ is equal to how many degrees?**
   A. 3 º B. 45 º C. 135 º D. 145 º
4. **What is 2.5 rev in degree?**
   A. 840º B. 860º C. 900º D. 920º
5. **An angle is said to be in standard position if the initial side of it lies on the _____.**
   A. 1st Quadrant B. +x - axis C. 2nd Quadrant D. –x - axis

Lesson 1 Angle in a Unit Circle

What's New



Angle is a plane figure formed by two rays that have common endpoint. The common end point is called *vertex*. Angle is defined as amount of space covered by the rotation of one of the rays. The fixed ray is called *initial side* while *terminal side* is the moving ray. The figure below illustrates an angle in standard position.

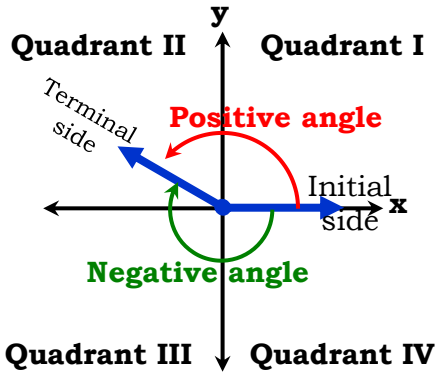
An **angle is in standard position** in the Cartesian plane when its initial side lies on the positive x-axis and the vertex is located at the origin. The location of the angle in the Cartesian plane is based on the last position of its terminal side.

In **plotting an angle in a rectangular coordinate plane**, placed the vertex on the origin and initial side on the positive x-axis. Then, rotate the terminal side either clockwise or counter-clockwise direction depending on the magnitude of the

given angle. For your convenience, you can also use the protractor in measuring the size of the angle.

Rotation of an Angle

An angle is the result of the rotation of its terminal side. The rotations are either clockwise or counterclockwise. A Counter-clockwise rotation is just the opposite direction of the movement of the clock while the clockwise rotation is similar to the movement of the hand of the clock. If the terminal side of an angle moves in ***counter-clockwise rotation***, the angle is in **positive direction** (+). If the terminal side of the angle moves in ***clockwise rotation***, the angle is in **negative direction** (–). The terminal side tells the quadrant where the angle lies.



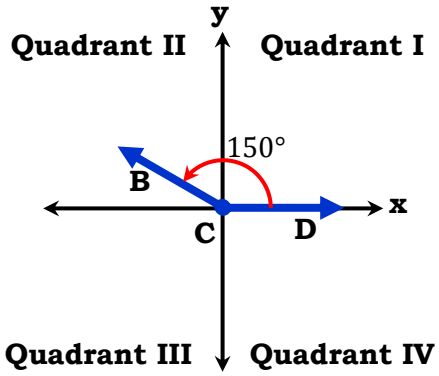
The four quadrants formed by the intersection of the x-axis and y-axis in the coordinate plane are numbered I, II, III, and IV counter-clockwise.

Take note that angle is a dimension thus, its value should not be negative. Negative sign pertains to its direction only.

Example:

$\angle DCB = 150°$ is in standard position. In what quadrant does the terminal side of $\angle DCB$ located.

$\angle DCB$ is located in quadrant II.



What is It

Angle Measurement

The measurement of the angle can be expressed in terms of Degree, Radian and Revolution. The focus of this module is to discuss the two most common angle units such as degree and radian.

1. *Degree* (°). It is an angle measurement taken from dividing a whole circle into 360 degree and one degree is equivalent to $\frac{1}{360}$ of a circle. A unit degree can be divided further into minute and second units. Thus, $1° = 60\ minutes$, 1 minute = 60 seconds.
2. *Radian (rad).* This is the standard unit of an angle. It is equivalent to the length of the arc or the circumference of a unit circle. It is the International Standard unit (SI) of an angle and commonly used by many professionals in the field of Mathematics. Radian is also defined as the ratio of the length of the arc and the radius of a given circle. A radian measures usually expressed as a fraction or multiple of π, **therefore a full unit circle equals 2π rad**.

Conversion of Angle Units

Conversion of angle measurements uses relationship between the two units that are about to be converted. Mathematical operations of multiplication and division are applied to process the conversion of the units.

- To convert the measure of an angle from degree to radian, multiply the given measure by $\frac{\pi\, rad}{180°}$.
- To convert the measure of an angle from radian to degree, multiply the given measure by $\frac{180°}{\pi\, rad}$.

Converting degrees to minutes, seconds

Convert 15.6° to: a) minutes; b) seconds

Solution:

Using the conversion factor:

$1° = 60'$,

a) 15.6° to minutes

$15.6°\ x\ \frac{60'}{1°} = 936'$

Solution:

$1° = 3600''$

b) 15.6° to seconds

$15.6°\ x\ \frac{3600"}{1°} = 56,160''$

Converting degrees to radians

1) Convert 120° to radians

Solution:

Using the conversion factor,

$120°\ x\ \frac{\pi\, rad}{180°} = \frac{2}{3}\pi$ radians

2) Convert 135° 22′16″ to radians

Solution:

$16" \left(\frac{1'}{60"}\right) = 0.267'$

$22.267" \left(\frac{1°}{60'}\right) = 0.371°$

$135^o + 0.371^o = 135.371°$

$135.371°\ x\ \frac{\pi\, rad}{180°} = \frac{3}{4}\pi$ radians

Converting radian to degrees

1) Convert $\frac{3}{4}\pi\ radians$ to degrees

Solution: $\frac{3\pi\, rad}{4}\ x\ \frac{180°}{\pi\, rad} = 135°$

2) Convert 2.23 radians to degree, minute, and second

Solution: Use 3.14 as value of π, → $\frac{180°}{\pi} = \frac{180°}{3.14}$

$= 57.32^o$

$2.23\ rad\ x\ \frac{57.32°}{rad} = 127.8236^o$

$0.8236°\ x\ \frac{60'}{1°} = 49.416'$

$0.416'\ x\ \frac{60"}{1'} = 24.96"$

2.23 radians =127° 49' 25"

What's More

A. Convert the following radian measures to Degree, Minute, Second (DMS) measures. Round off seconds to nearest whole number if necessary.

1) 1.07 rad
2) 2.28 rad
3) 1.015 rad
4) 2. 36 rad
5) 1.005 rad

B. Convert the following degree measures to Degree, Minute, Second (DMS) measures.

1) 225.48°
2) 15.24°
3) 308.12°
4) 112.25°
5) 620.18°

What I can Do

The following angles are in standard position. Draw each angle and determine the quadrant where the terminal side lie. Use a protractor to draw the angle with correct measurement.

1) $\angle ABC = 85°$ 3) $\angle MAT = 234°$ 5) $\angle RAY = -297°$
2) $\angle ATE = 167°$ 4) $\angle CAB = -72°$

Assessment

General Directions: Convert the following. Avoid erasures.

I. Convert the following degree measures to radian measures.

1) 75° 3) 36° 5) 159° 7) 416° 9) 250°
2) 228° 4) 378° 6) 45° 8) 108° 10) 270°

II. Convert the following radian measures to degree measures.

1) $\frac{2}{5}\pi\ rad$ 3) $\frac{15}{4}\pi\ rad$ 5) $\frac{12}{7}\pi\ rad$ 7) $\frac{9}{4}\pi\ rad$ 9) $\frac{8}{5}\pi\ rad$
2) $\frac{3}{7}\pi\ rad$ 4) $\frac{4}{9}\pi\ rad$ 6) $\frac{11}{4}\pi\ rad$ 8) $\frac{1}{6}\pi\ ra$ 10) $\frac{10}{3}\pi\ rad$

What I Need to Know

Expected Learning Outcome: **(LC Codes: STEM_PC11AG-IIb-1)**

1. Illustrate angles in standard position and coterminal angles.

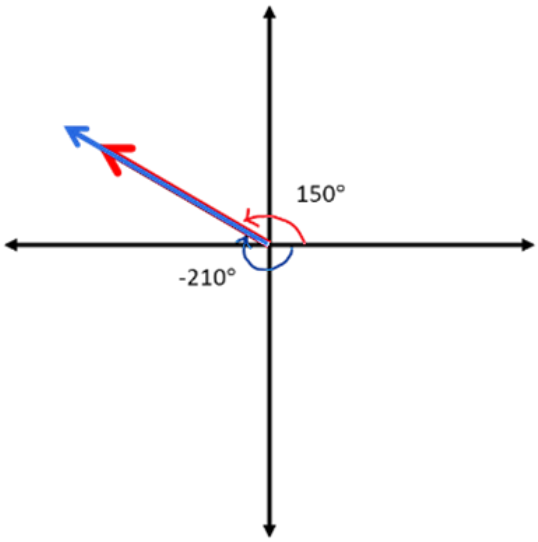
What I know

Directions: Write the letter of the correct answer on a separate sheet of paper.

1) What do you call the two angles in standard position with the same terminal side?
   A. coterminal angle C. special angle
   B. reference angle D. quadrantal angle
2) What do you call the acute angle formed by the terminal side of a given angle and the x-axis?
   A. coterminal angle C. special angle
   B. reference angle D. quadrantal angle
3) Which of the following is co-terminal with –120°?
   A. 240° B. 250° C. –240° D. 120°
4) The following are co terminal angles except what?
   A. π/6 B. 390° C. -330° D. π/4
5) In which quadrant does the terminal side of $\angle SET = -494°$ lie?
6) A. I B. II C. III D. IV

Lesson 2 Coterminal Angles

What's New



It is possible for more than one angle to have the same terminal side. Look at the figure at the right. The angle 150° is a positive angle whose terminal side rotated counterclockwise. The angle −210° is a negative angle whose terminal side rotated clockwise but both angles have common terminal sides. If two angles in standard position have the same terminal side, they are said to be ***coterminal angles***.

Every angle greater than 360° or less than 0° has a coterminal angle with measurement between 0° and 360°, and it is often more convenient to find the coterminal angle within the range of 0° to 360° than to work with an angle that is outside that range.

Any angle has infinitely many coterminal angles because adding or subtracting multiples of 360° or 2π to that angle, the resulting angle has the same terminal side to the given angle.

Examples:

100°, –260°, –620°, 460°, and 820° are coterminal because,

$100^\circ - 360^\circ = -260^\circ$, $\quad 100^\circ - 720^\circ = -620^\circ$

$100^\circ + 360^\circ = 460^\circ$, $\quad 100^\circ + 720^\circ = 820^\circ$

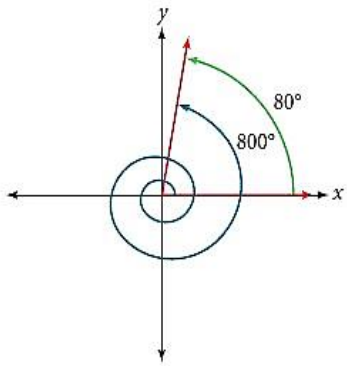
What is It

To find a coterminal angle (from 0° to 360°) of an angle greater than 360° or less than 0°, subtract or add, respectively a multiple of 360° that will give a difference within the range of 0° to 360°. It is the same process with subtracting or adding 360° repeatedly from the given angle until the result is within the range of 0° to 360°.

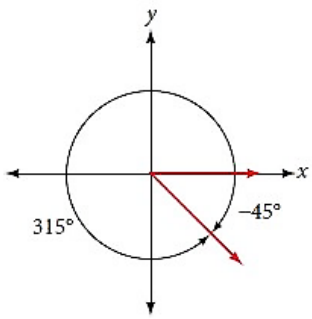
Similar to degree measure, given a radian measure of an angle greater than 2π or less than 0 radian, subtract or add, respectively a multiple of 2π that will give a difference within the range of 0 to 2π radian. It is also the same process with

subtracting or adding 2π repeated from the given angle until the result is within the range of 0 to 2π radian.

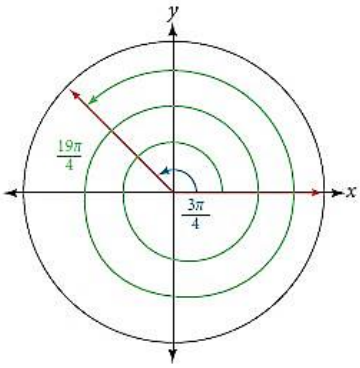
1. Find the least positive angle θ that is coterminal with an angle measuring 800°, where $0° \leq \theta < 360°$.
   Solution: An angle with measure 800° is coterminal with an angle with measure 800 − 360 = 440°, but 440° is still greater than 360°, so we subtract 360° again to find another coterminal angle: 440 − 360 = 80°. The angle θ = 80° is coterminal with 800°.
   Alternate Solution: 800° is coterminal with 800° – 2(360°) = 800° – 720° = 80°.



2. Show the angle with measure −45° on a circle and find a positive coterminal angle θ such that $0° \leq \theta < 360°$.
   Solution: $\theta + 360n$
   $= -45° + 360(1)$
   $= -45° + 360°$
   $= 315°$



3. Find an angle θ that is coterminal with $\frac{19\pi}{4}$, where $0 \leq \theta < 2\pi$.
   Solution: $\theta - (2\pi)n$
   $= \frac{19\pi}{4} - (2\pi)(2)$
   $= \frac{19\pi}{4} - 4\pi$
   $= \frac{3\pi}{4}$



What's More

A. Find the smallest positive coterminal angle of the following.

1) 718°
2) 983°
3) 460°
4) –436°
5) –867°
6) $\frac{3\pi}{5}$
7) $\frac{13\pi}{4}$
8) $\frac{22\pi}{7}$
9) $-\frac{7\pi}{3}$
10) $-\frac{4\pi}{9}$

B. Find the largest negative coterminal angle of the following.

1) 314°
2) 807°
3) 501°
4) –109°
5) –1 098°
6) $\frac{13\pi}{4}$
7) $\frac{2\pi}{3}$
8) $\frac{27\pi}{5}$
9) $-\frac{7\pi}{11}$
10) $-\frac{9\pi}{2}$

What I can Do

Determine if each pair of the two angles are co terminal angles. If it is co terminal angles, the angles in a Cartesian plane if NOT co terminal, with NC.

1) 185°, 545° 2) $\frac{17\pi}{36}, \frac{161\pi}{36}$ 3) 225°, –135°

Find a coterminal angle between 0° and 360° for each given angle.

1) –330° 2) 640° 3) –435° 4) –442° 5) 405°

Find a coterminal angle between 0 and 2π for each given angle.

1) $\frac{11\pi}{3}$ 2) $-\frac{35\pi}{18}$ 3) $\frac{15\pi}{4}$ 4) $-\frac{19\pi}{12}$ 5) $\frac{20\pi}{7}$

Find a positive and a negative coterminal angle for each given angle.

1) $\frac{5\pi}{4}$ 2) $\frac{25\pi}{36}$ 3) 373° 4) 178° 5) –567°

Assessment

Multiple Choice: Choose the letter of the correct answer.

___ 1. Which one is not a coterminal angle of 130°?

A. 490° B. -230° C. $\frac{23\pi}{18}$ D. $-\frac{23\pi}{18}$

___ 2. Which of the following angles does NOT belong to the group?

A. 60° B. -120° C. $\frac{4\pi}{3}$ D. $-\frac{2\pi}{3}$

___ 3. What is the smallest positive coterminal angle of 945°?

A. 315° B. 335° C. $\frac{4\pi}{5}$ D. $\frac{5\pi}{4}$

___ 4. What is the largest negative coterminal angle of $-\frac{11\pi}{3}$?

A. -120° B. -240° C. $-\frac{5\pi}{3}$ D. $-\frac{17\pi}{3}$

___ 5. Three angles are coterminal angles EXCEPT

A. 30° B. 190° C. - 330° D. 390°

What I Need to Know

Expected Learning Outcomes: **(LC Codes: STEM_PC11AG-IIc-1-2)**

1. Illustrate the different circular functions; and
2. Use reference angles to find the exact values of circular functions.

What I know

A. Write the letter of the correct answer on a separate sheet of paper.

1. **The reciprocal relation of secant function is _____.**
   A. sine function C. cosecant function
   B. cosine function D. cotangent function
2. **What is the correct way to represent the position of angle X on a circle as an ordered pair (x,y)?**
   A. (sinX, cosX) B. (cosX, sinX) C. (rsinX, rcosX) D. (rcosX, rsinX)

B. Angle θ is in standard position whose terminal side intersects the circle with radius r at point (x, y). Write the letter of the ratio in column B that is equivalent to the function in column A.

COLUMN A	COLUMN B
1) $cos\ \theta$	A. $\frac{x}{y}$
2) $tan\ \theta$	B. $\frac{r}{y}$
3) $cot\ \theta$	C. $\frac{x}{r}$
4) $sec\ \theta$	D. $\frac{y}{x}$
5) $csc\ \theta$	E. $\frac{r}{x}$

Lesson 3 Circular Function

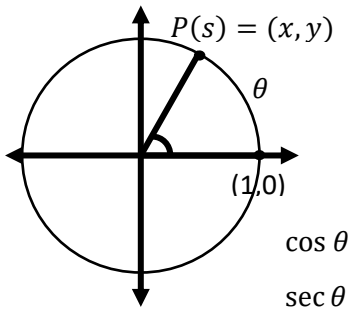
What's New

Circular Functions on Real Numbers

For any real numbers, we can find the values of the six circular functions using the ***unit circle***, a circle with radius 1 having the equation $x^2 + y^2 = 1$. Starting the point (1,0), we move $|s|$ units along the circumference of the circle, counterclockwise if $s > 0$ and clockwise if $s < 0$. We let $P(s)$ be the terminal point of arc s. If $P(s)$ is the point (x, y) on the unit circle then we define the six circular functions as follows.

$$\cos s = x \qquad \sin s = y \qquad \tan s = \frac{y}{x}, x \neq 0$$

$$\sec s = \frac{1}{x} \qquad \csc s = \frac{1}{y} \qquad \cot s = \frac{x}{y}, y \neq 0$$



We let s be the arclength subtended by angle θ which is in standard position on the unit circle. We have learned that $s = r\theta$. Since $r = 1$, then $s = \theta$. This means that θ has radian measure equal s. Using the definition of the six circular functions, we have

$\cos\theta = \frac{x}{r} = \frac{x}{1} = x$ $\quad$ $\sin\theta = \frac{y}{r} = \frac{y}{1} = y$ $\quad$ $\tan\theta = \frac{y}{x}$

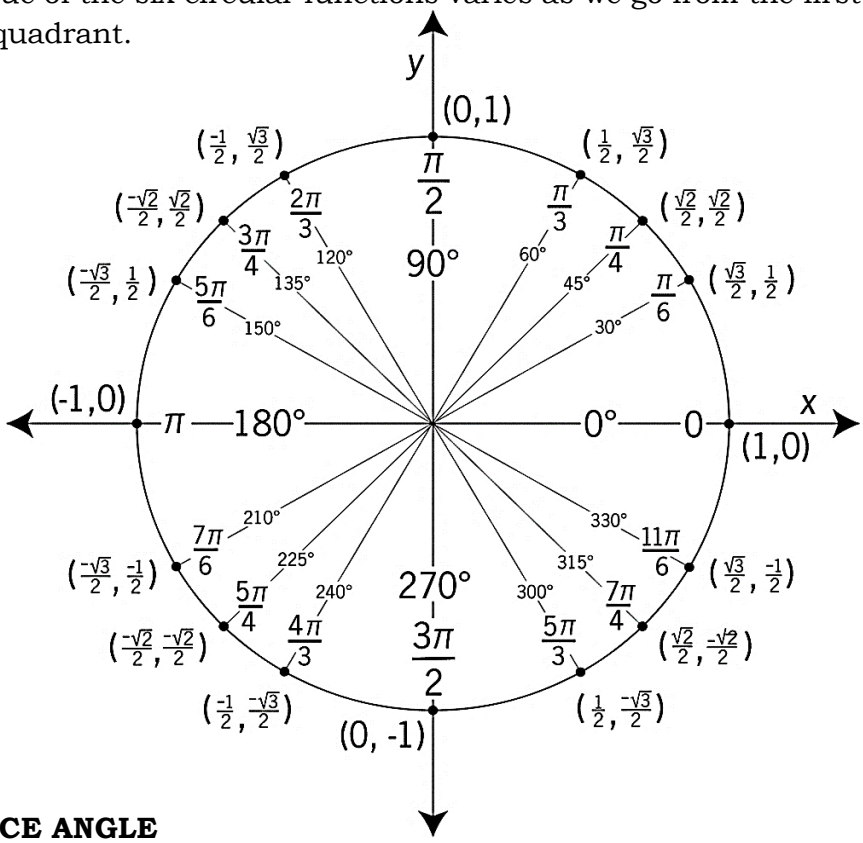
$\sec\theta = \frac{1}{x}$ $\quad$ $\csc\theta = \frac{1}{y}$ $\quad$ $\cot\theta = \frac{x}{y}$

Clearly, we can see that the values of the trigonometric functions and its corresponding circular functions are the same.

What is It

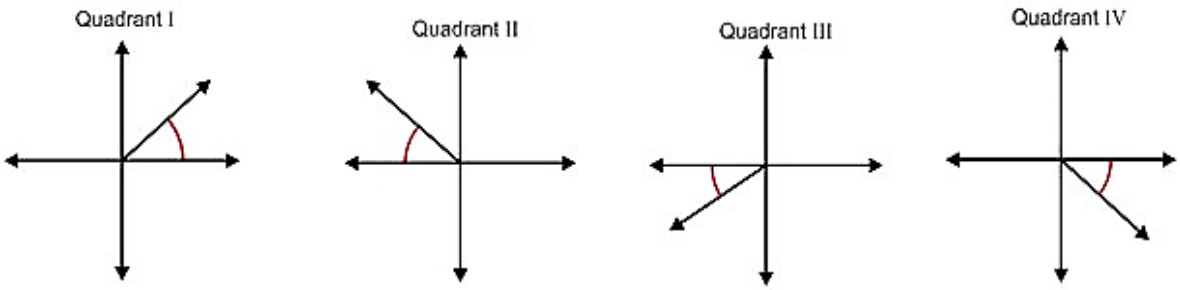
The Terminal Point $P(\theta)$ of Some Special Angles

The value of the six circular functions varies as we go from the first quadrant to the fourth quadrant.



REFERENCE ANGLE

- The smallest positive acute angle between the x-axis and the terminal side



Finding the Reference Angle

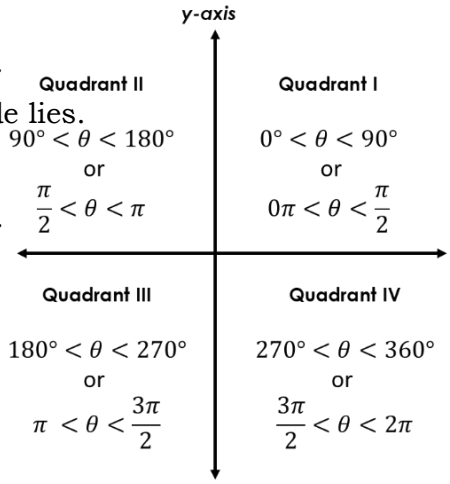
	I	II	III	IV
$\mathbf{0° \le \theta \le 360°}$	same as θ	$180° - \theta$	$\theta - 180°$	$360° - \theta$
$\mathbf{0\pi \le \theta \le 2\pi}$		$\pi - \theta$	$\theta - \pi$	$2\pi - \theta$

For $\theta < 0°$ or $\theta < 0\,radian$, and $\theta > 360°$ or $\theta > 2\pi$

Step 1: Find the smallest positive coterminal angle.

Step 2: Determine which quadrant the terminal side lies.

Step 3: Find the reference angle using the table above mentioned for $0° \le \theta \le 360°$ or for $0\pi \le \theta \le 2\pi$.



Find the exact value of each of the following.

1) $\boldsymbol{cos\ 240°}$ ***Solution:***

Step 1: Identify which quadrant 240° lies. → 240° is in **Quadrant III**

Step 2: Find the reference angle of 240°. Reference Angle: $\theta - 180°$
$= 240° - 180°$
$\mathbf{= 60°}$

Step 3: Determine the coordinates of the terminal point. $P(60°) = \left(\frac{1}{2}, \frac{\sqrt{3}}{2}\right)$

Note: *Since the given angle is in Quadrant III, the sign of coordinates must be* $(-,-)$, *so* $\left(-\frac{1}{2}, -\frac{\sqrt{3}}{2}\right)$

Step 4: Definition of circular function. $\cos\theta = x$

$$\boxed{\boldsymbol{cos\ 240° = -\frac{1}{2}}}$$

2) $\boldsymbol{sin\ 690°}$ ***Solution:***

Step 1: Find the smallest positive coterminal angle of 690° → $\theta - 360°n$
$690 - 360°(1) = \mathbf{330°}$

Step 2: Identify which quadrant 330° lies. 330° is in **Quadrant IV**

Step 3: Find the reference angle of 330°. Reference Angle: $360° - \theta$
$= 360° - 330°$
$\mathbf{= 30°}$

Step 4: Determine the coordinates of the terminal point → $P(30°) = \left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$

Note: *Since the given angle is in Quadrant IV the sign of coordinates must be* $(+,-)$, *so* $\left(\frac{\sqrt{3}}{2}, -\frac{1}{2}\right)$

Step 5: Definition of circular function. $\sin\theta = y$

$$\boxed{\boldsymbol{sin\ 690° = -\frac{1}{2}}}$$

3) $\boldsymbol{sec\ (-405°)}$ ***Solution:***

Step 1: Find the smallest positive coterminal angle of 690° → $\theta + 360°n$
$-405° - 360°(2) = \mathbf{315°}$

Step 2: Identify which quadrant 315° lies. 315° is in **Quadrant IV**

Step 3: Find the reference angle of 315°. Reference Angle: $360° - \theta$
$360° - 315° = \mathbf{45°}$

Step 4: Determine the coordinates of the terminal point. $P(45°) = \left(\frac{\sqrt{2}}{2}, \frac{\sqrt{2}}{2}\right)$

Note: *Since the given angle is in Quadrant IV the sign of coordinates must be* $(+,-)$. *So,* $\left(\frac{\sqrt{2}}{2}, -\frac{\sqrt{2}}{2}\right)$

Step 5: Definition of circular function. $\sec\theta = \frac{1}{x}$

$$\sec(-405°) = \frac{1}{\frac{\sqrt{2}}{2}}$$

$$= \frac{1}{\frac{\sqrt{2}}{2}} = 1 \cdot \frac{2}{\sqrt{2}} = \frac{2}{\sqrt{2}}$$

rationalize and simplify $\frac{2}{\sqrt{2}} \cdot \frac{\sqrt{2}}{\sqrt{2}} = \frac{2\sqrt{2}}{\sqrt{4}} = \frac{2\sqrt{2}}{2} = \sqrt{2}$

$$\boxed{\boldsymbol{sec}\,(-405°) = \sqrt{2}}$$

4. $\boldsymbol{csc\left(\frac{17\pi}{6}\right)}$ ***Solution:***

Step 1: Find the smallest positive coterminal angle of 690° → $\theta - 2\pi n$

$$\frac{17\pi}{6} - 2\pi°(1) = \frac{5\pi}{6}$$

Step 2: Identify which quadrant $\frac{5\pi}{6}$ lies. 315° is in **Quadrant II**

Step 3: Find the reference angle of $\frac{5\pi}{6}$. Reference Angle: $\pi - \theta$

$$= \pi - \frac{5\pi}{6}$$

$$= \frac{\pi}{6}$$

Step 4: Determine the coordinates of the terminal point. $P\left(\frac{\pi}{6}\right) = \left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$

Note: *Since the given angle is in Quadrant II the sign of coordinates must be* $(-,+)$. *So,* $\left(-\frac{\sqrt{3}}{2}, \frac{1}{2}\right)$

Step 5: Definition of circular function. $\csc\theta = \frac{1}{y}$

$$\sec\left(\frac{17\pi}{6}\right) = \frac{1}{\frac{1}{2}}$$

$$= \frac{1}{\frac{1}{2}} = 1 \cdot \frac{2}{1} = 2$$

$$\boxed{\boldsymbol{sec}\left(\frac{17\pi}{6}\right) = 2}$$

What's More

Evaluate each of the following.

1) $\tan(-135°)$
2) $sec\left(-\frac{5\pi}{3}\right)$
3) $csc\ \frac{23\pi}{6}$
4) $cot\ 510°$
5) $sin\ 90° + \cos 120°$
6) $sin\ 120° - \cos 60°$
7) $cos\ \frac{5\pi}{3} - \tan\frac{7\pi}{4}$
8) $csc\ \frac{5\pi}{6} + \cot\frac{9\pi}{4}$
9) $\sec\frac{11\pi}{6} \cdot \cos 5\pi$
10) $\sec\frac{\pi}{4} + \cot\frac{7\pi}{3}$

What I can Do

Directions: Solve the following problems.

1) If $P(\theta)$ is a point on the unit circle and $\theta = \frac{23\pi}{3}$, what is the coordinates of $P(\theta)$?

2) If $P(\theta)$ is a point on the unit circle and $\theta = -\frac{17\pi}{6}$, find the values of the six trigonometric functions of θ.

Assessment

Directions: Find the values of the six trigonometric functions for the given θ.

θ	$\sin\theta$	$\cos\theta$	$\tan\theta$	$\csc\theta$	$\sec\theta$	$\cot\theta$
$240°$						
$-315°$						
$\frac{4\pi}{3}$						
$\frac{11\pi}{4}$						
$-\frac{20\pi}{8}$						

What I Need to Know

Expected Learning Outcomes: **(LC Codes: STEM_PC11AG-IId-1-2)**

1. Illustrate the domain and range of the different circular functions; and
2. Graph the six circular functions (a) amplitude, (b) period, and (c) phase shift.

What I know

Write the letter of the correct answer on a separate sheet of paper.

Choices for items 1 – 4

A. Amplitude B. Period C. Maximum point D. Minimum point

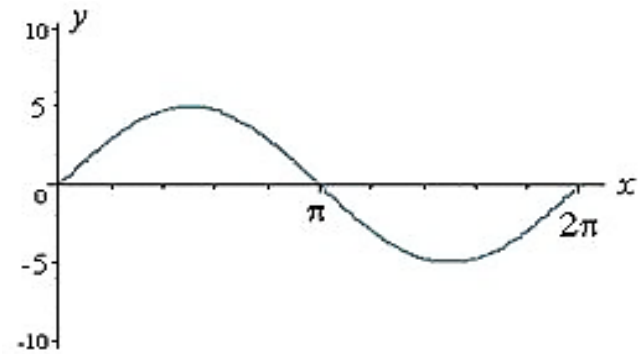
_____ 1. This refers to the length of one complete cycle.

_____ 2. It is the highest point of a graph above x – axis.

_____ 3. It is the portion of the graph where it starts to repeat itself.

_____ 4. It indicates the height of the graph above or below the x – axis.

For items 5 - 6, please refer to figure below.



_____ 5. What is the equation of the graph?
A. y = 5 sin 2x
B. y = 5 cos x
C. y = 5 sin x
D. y = 5 cos 2x

_____ 6. What is the amplitude?
A. 1 C. 4
B. 2 D. 5

Lesson 4 Graph of Circular Function

What's New

Recall that the sine and cosine functions relate real number values to the *x*- and *y*-coordinates of a point on the unit circle, so what do they look like on a graph on a coordinate plane? Let's start with the sine function. We can create a table of values and use them to sketch a graph. **Table 1** lists some of the values for the sine function on a unit circle.

x	**0**	$\frac{\pi}{6}$	$\frac{\pi}{4}$	$\frac{\pi}{3}$	$\frac{\pi}{2}$	$\frac{2\pi}{3}$	$\frac{3\pi}{4}$	$\frac{5\pi}{6}$	π
sin *x*	0	$\frac{1}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{\sqrt{3}}{2}$	1	$\frac{\sqrt{3}}{2}$	$\frac{\sqrt{2}}{2}$	$\frac{1}{2}$	0

Table 1

Plotting the points from the table and continuing along the *x*-axis gives the shape of the sine function. See **Figure 2**.

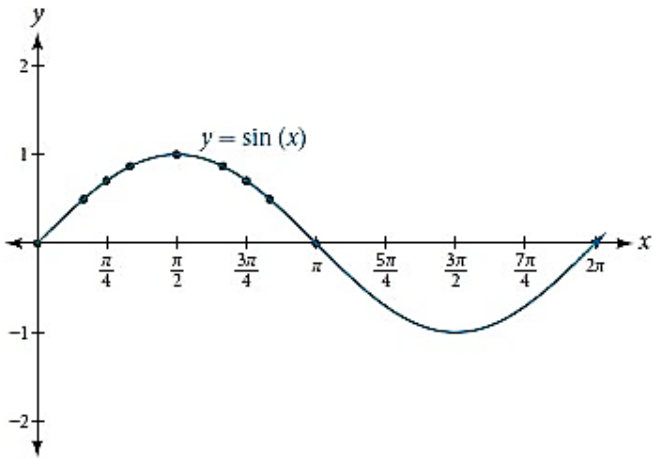
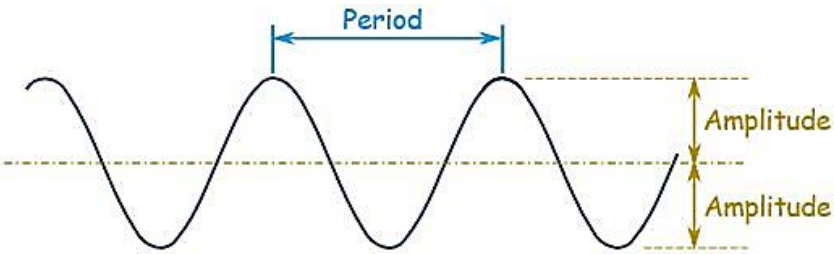


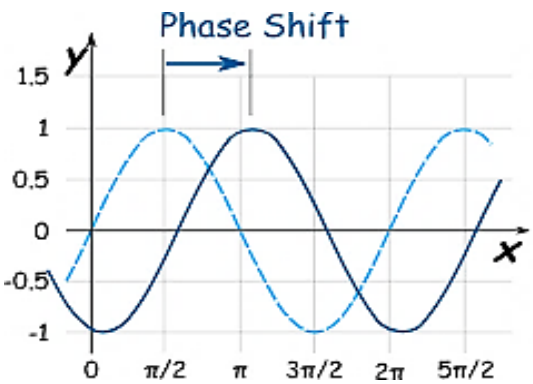
Figure 2 The sine function

- **The period** is the length of one complete cycle.

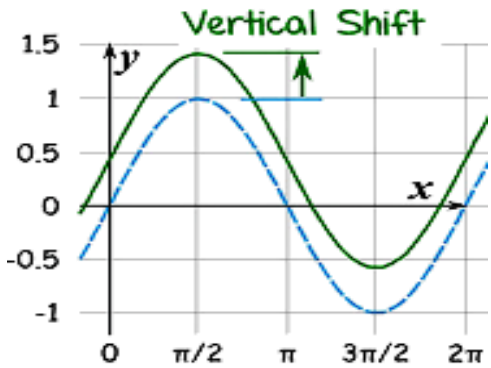


- The **Amplitude** is the height from the center line to the peak (or to the trough). We can measure the height from highest to lowest points and divide that by 2.

- The **Phase Shift** is how far the function is shifted **horizontally** from the usual position.

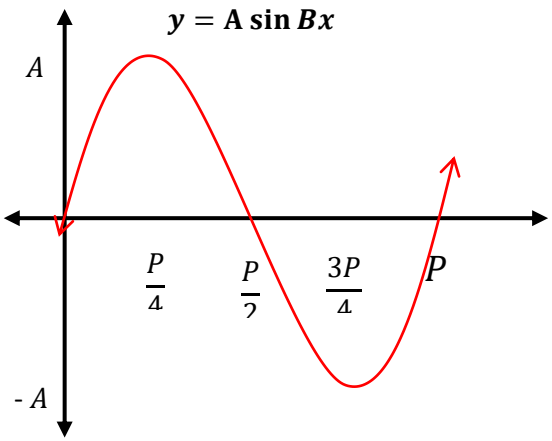


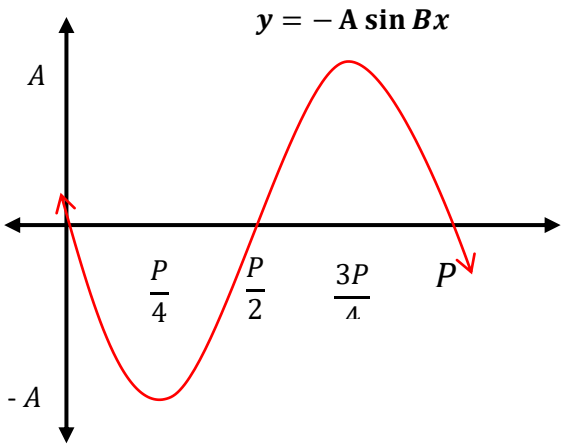
- The **Vertical Shift** is how far the function is shifted **vertically** from the usual position.



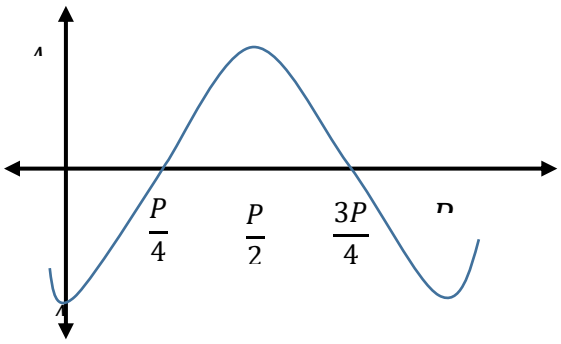
What is It

The Graph of Circular Functions when C=0 and D=0

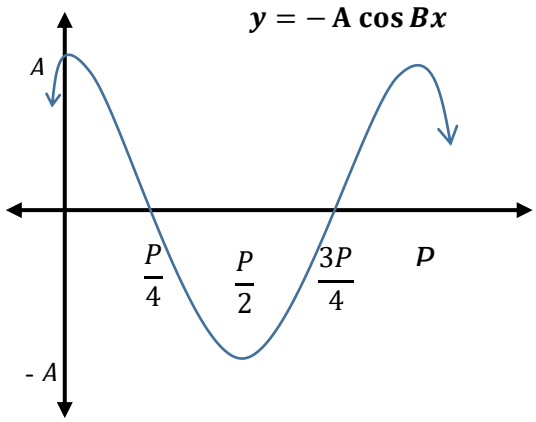




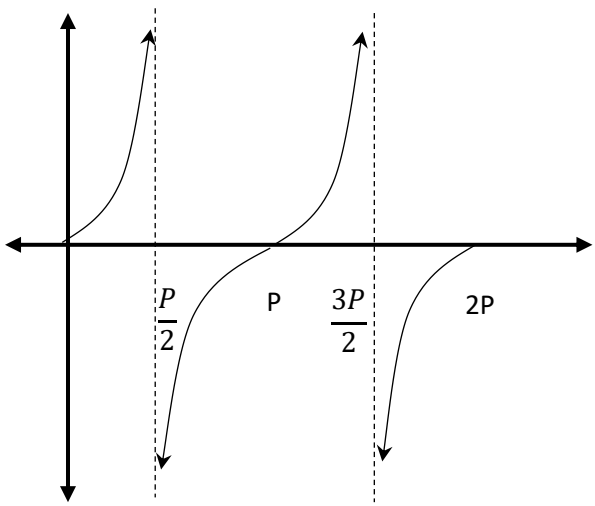
$y = \mathbf{A}\cos Bx$



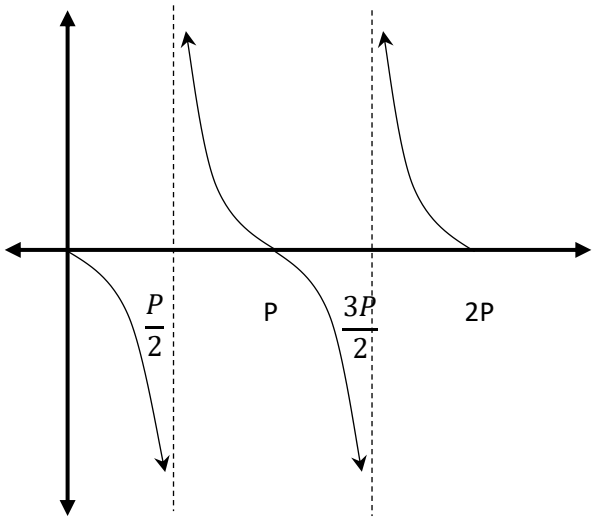
$y = -\mathbf{A}\cos Bx$



$y = \mathbf{A}\tan Bx$



$y = -\mathbf{A}\tan Bx$



Characteristics of Sine and Cosine Functions

The sine and cosine functions have several distinct characteristics:

- Both are periodic functions with a period of $\frac{2\pi}{|B|}$.
- The domain of each function is $(-\infty, \infty)$ and the range is $[-A, A]$.
- The graph of $y = \sin x$ is symmetric about the origin, because it is an odd function.
- The graph of $y = \cos x$ is symmetric about the y- axis, because it is an even function.

Different Equations, Same Graph

1) The graphs of $y = \sin x$ and $y = \sin(x + 2\pi k)$, k is any integer, are the same.
2) The graphs of $y = \sin x$, $y = -\sin(x + \pi)$, $y = \cos\left(x - \frac{\pi}{2}\right)$, and $= -\cos\left(x + \frac{\pi}{2}\right)$ are the same.

3) In general, the graphs of

$y = \mathrm{A}\sin B\,(x - C) + D$,

$y = -\,\mathrm{A}\sin[\,B(x - C) + \pi + 2\pi k] + D$,

$y = \mathrm{A}\cos\left[\,B(x - C) - \frac{\pi}{2} + 2\pi k\right] + D$, and

$y = -\,\mathrm{A}\cos\left[\,B(x - C) + \frac{\pi}{2} + 2\pi k\right] + D$,

where k is any integer, are all the same.

Similar observations are true for cosine.

Graph each of the following.

1. $f(x) = \sin x$
2. $f(x) = -4\sin 8x$
3. $f(x) = 2\cos x$
4. $f(x) = -3\cos 2x$
5. $f(x) = \tan 4x$

Solution:

1) Given a function in the form $f(x) = Asin(Bx - C) + D$, identify the midline, amplitude, period, and phase shift.

   a. Determine the amplitude as $|A|$. → Ans. $|\boldsymbol{A}| = \mathbf{1}$

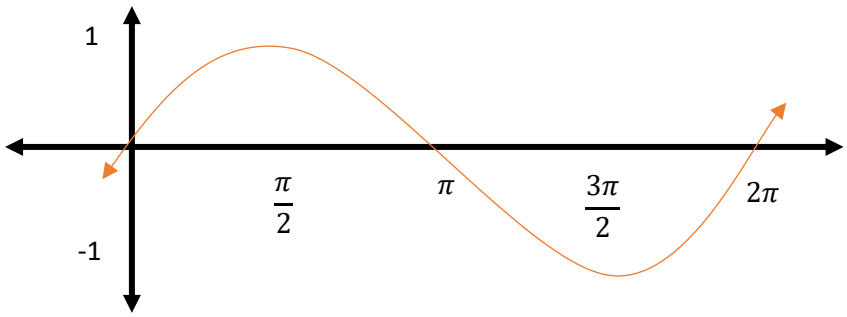
   b. Determine the period as $P = \frac{2\pi}{|B|}$ → $P = \frac{2\pi}{|B|} = \frac{2\pi}{|1|} = \boldsymbol{2\pi}$

   Divide the period ($P = \boldsymbol{2\pi}$) into 4 equal parts.

   $P \cdot \frac{1}{4} = 2\pi \cdot \frac{1}{4} = \boldsymbol{\frac{\pi}{2}}$ $\quad P \cdot \frac{1}{2} = 2\pi \cdot \frac{1}{2} = \boldsymbol{\pi}$ $\quad P \cdot \frac{3}{4} = 2\pi \cdot \frac{3}{4} = \boldsymbol{\frac{3\pi}{2}}$

   c. Determine the phase shift as $\frac{C}{B}$ → phase shift $= \frac{C}{B} = \frac{0}{1} = \mathbf{0}$

   d. Determine the midline as $y = D$. → $y = D = \mathbf{0}$



2. $\boldsymbol{f(x) = 4\sin 8x}$

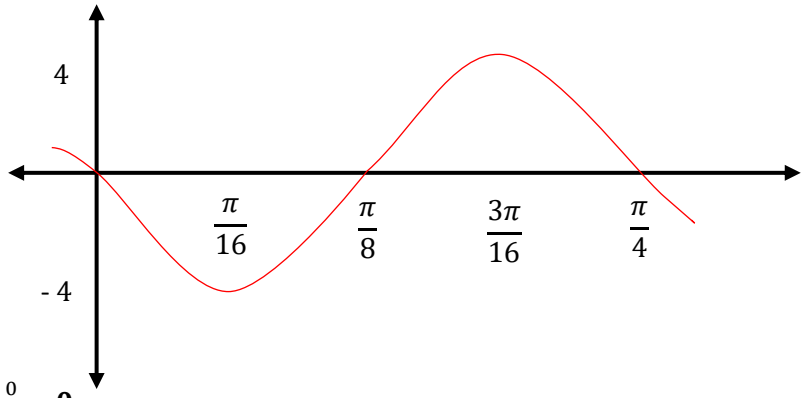
   a. $|A| = |-4| = \mathbf{4}$

   b. $P = \frac{2\pi}{|B|} = \frac{2\pi}{|8|} = \boldsymbol{\frac{\pi}{4}}$

$P \cdot \frac{1}{4} = \frac{\pi}{4} \cdot \frac{1}{4} = \boldsymbol{\frac{\pi}{16}}$

$P \cdot \frac{1}{2} = \frac{\pi}{4} \cdot \frac{1}{2} = \boldsymbol{\frac{\pi}{8}}$

$P \cdot \frac{3}{4} = \frac{\pi}{4} \cdot \frac{3}{4} = \boldsymbol{\frac{3\pi}{16}}$



   c. phase shift $= \frac{C}{B} = \frac{0}{8} = \mathbf{0}$

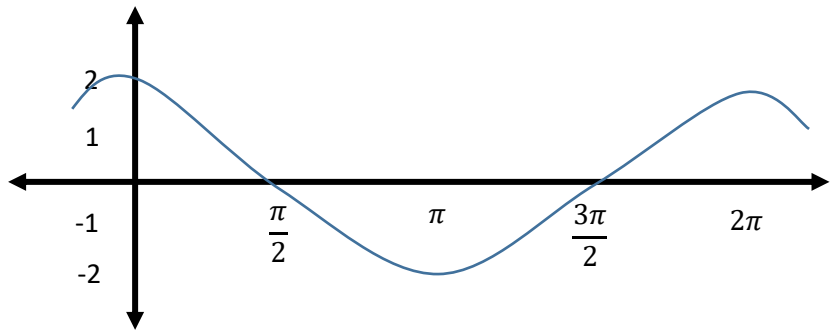
   d. $y = D = \mathbf{0}$

3. $\boldsymbol{f(x) = 2\cos x}$

   a. $|A| = |2| = \mathbf{2}$

   b. $P = \frac{2\pi}{|B|} = \frac{2\pi}{|1|} = \boldsymbol{2\pi}$

- $P \cdot \frac{1}{4} = 2\pi \cdot \frac{1}{4} = \boldsymbol{\frac{\pi}{2}}$
- $P \cdot \frac{1}{2} = 2\pi \cdot \frac{1}{2} = \boldsymbol{\pi}$
- $P \cdot \frac{3}{4} = 2\pi \cdot \frac{3}{4} = \boldsymbol{\frac{3\pi}{2}}$



   c. phase shift $= \frac{C}{B} = \frac{0}{8} = \mathbf{0}$

   d. $y = D = \mathbf{0}$

4. $\boldsymbol{f(x) = -3\cos 2x}$

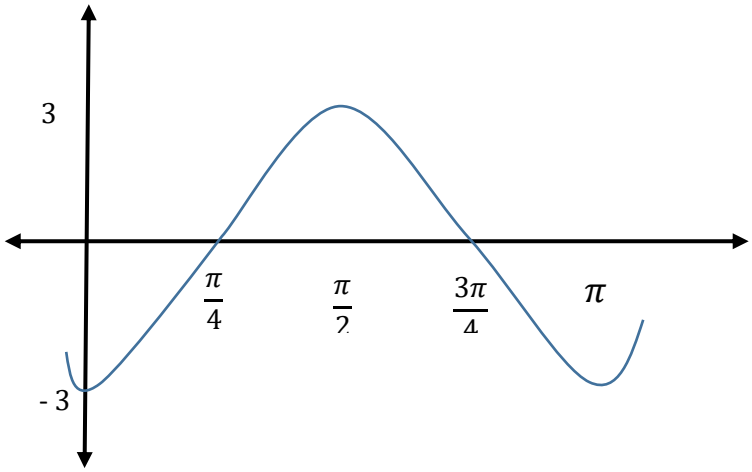
a. $|A| = |-3| = \mathbf{3}$

b. $P = \frac{2\pi}{|B|} = \frac{2\pi}{|2|} = \boldsymbol{\pi}$

- $P \cdot \frac{1}{4} = \pi \cdot \frac{1}{4} = \boldsymbol{\frac{\pi}{4}}$
- $P \cdot \frac{1}{2} = \pi \cdot \frac{1}{2} = \boldsymbol{\frac{\pi}{2}}$
- $P \cdot \frac{3}{4} = \pi \cdot \frac{3}{4} = \boldsymbol{\frac{3\pi}{4}}$

c. phase shift = $\frac{C}{B} = \frac{0}{8} = \mathbf{0}$

d. $y = D = \mathbf{0}$

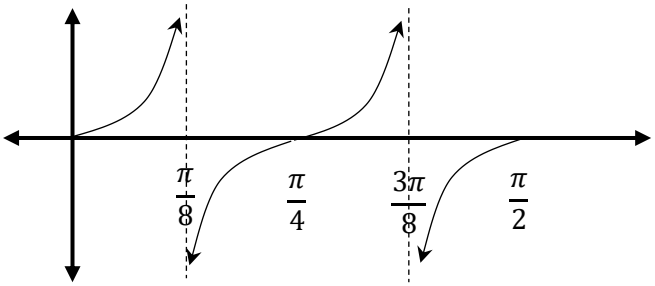


5. $\boldsymbol{f(x) = f(x) = \tan 4x}$

a. Determine the period. → $P = \frac{\pi}{|B|} = \frac{\pi}{|4|} = \boldsymbol{\frac{\pi}{4}}$

b. Determine the two adjacent vertical asymptotes. → $\boldsymbol{P \cdot \frac{1}{2}} = \frac{\pi}{4} \cdot \frac{1}{2} = \boldsymbol{\frac{\pi}{8}}$

$\boldsymbol{P \cdot \frac{3}{2}} = \frac{\pi}{4} \cdot \frac{3}{2} = \boldsymbol{\frac{3\pi}{8}}$



What's More

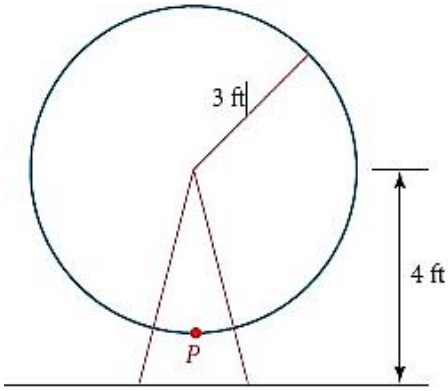
Graph the following equations. In your graph, show <u>only one</u> period of the function.

1) $f(x) = -2\sin x$
2) $f(x) = 3\sin 3x$
3) $f(x) = 2\cos\frac{1}{2}x$
4) $f(x) = -\frac{1}{4}\cos 5x$
5) $f(x) = -\tan 2x$

What I can Do

Solve the given problem.

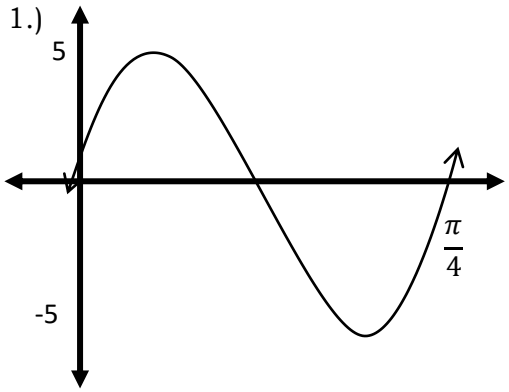
A circle with radius 3 ft is mounted with its center 4 ft off the ground. The point closest to the ground is labeled *P*, as shown in the figure. Sketch a graph of the height above the ground of the point *P* as the circle is rotated; then find a function that gives the height in terms of the angle of rotation.



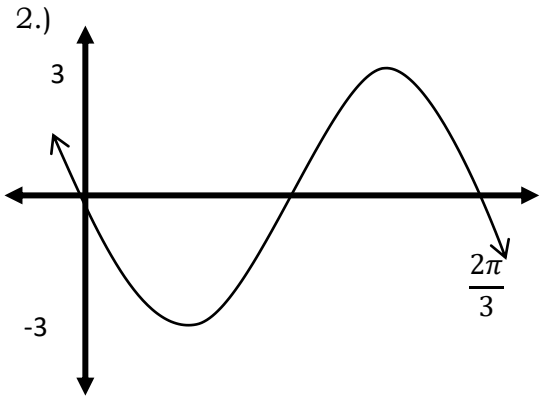
Assessment

Directions: Determine the (a) amplitude, (b) period, (c) midline, and an (d) equation involving sine and cosine for the graph shown in the figures.

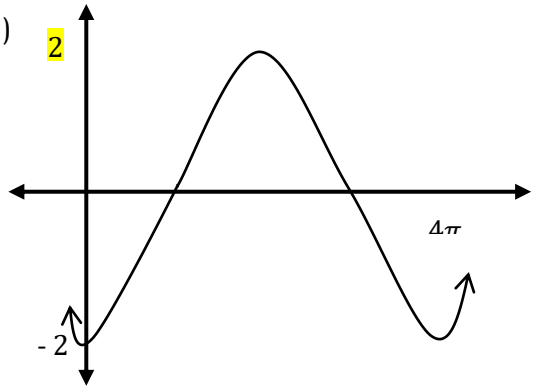
1.)



2.)



3.)



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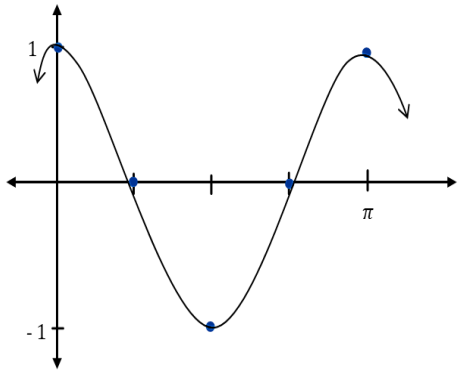
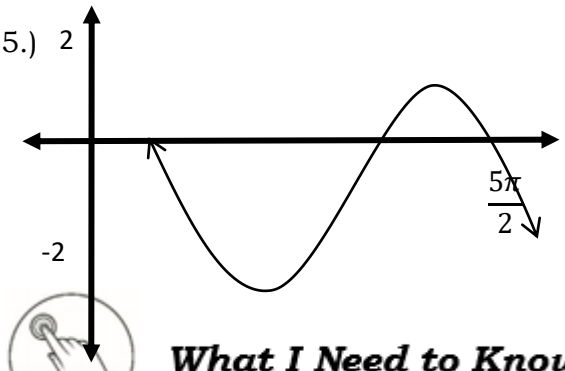


5.)



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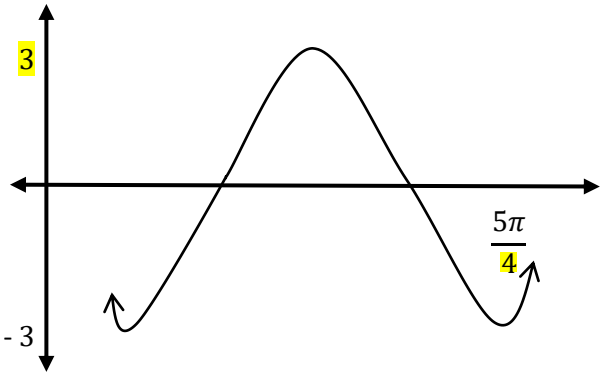


What I Need to Know

Expected Learning Outcome: **(LC Codes: STEM_PC11AG-IIe-1)**

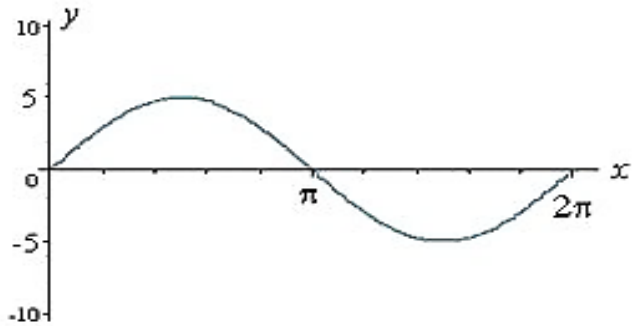
1. Solve situational problems involving circular functions.

What I know

Write the letter of the correct answer on a separate sheet of paper.
For numbers 1- 3, please refer to the figure on the next page.

_____ 1. **What is the equation of the graph?**

A. y = 5 sin 2x
B. y = 5 cos x
C. y = 5 sin x
D. y = 5 cos 2x



_____ 2. **What is the amplitude?**

A. 1 C. 4
B. 2 D. 5

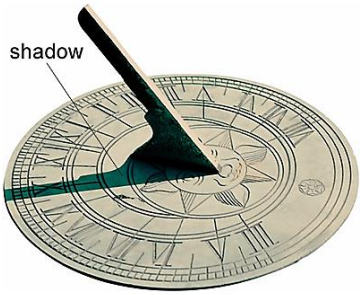
_____ 3. **What is the period?**

A. π B. 2π C. π/2 D. 5 π

Lesson 5 — Problems Involving Circular Functions

What's New

SIMPLE HARMONIC MOTION



Repetitive or periodic behavior is common in nature. As an example, the time-telling device known as *sundial* is a result of the predictable rising and setting of the sun every day. It consists of a flat plate and a gnomon. As the sun moves across the sky, the gnomon casts a shadow on the plate, which is calibrated to tell the time of the day.

Some motions are also periodic. When a weight is suspended on a spring, pulled down, and released, the weight oscillates up and down. Neglecting resistance, this oscillatory motion of the weight will continue and on, and its height is periodic with respect to time.

What is It

Now that we know the **Unit Circle** inside out, let us **graph the circular functions** on the coordinate system. The x- values are the angles (in radians – that is the way it is done); and the y- values are the circular values (like **sin**, **cos**, and **tan**).

The **sin** and **cos** (and **csc** and **sec**) functions start repeating after 2π radians, and it turns out that the **tan** and **cot** functions start repeating again after only π radians. The reason **tan** (and **cot)** repeat after only π radians is because, when dividing **sin** and **cos** to get **tan**, we get the same values in **quadrants III** as **IV** as we do for **quadrants I** and **II**. (Try this with the Unit Circle).

A complete repetition of the pattern of the function is called a **cycle** and the **period** is the horizontal length of one complete cycle. Thus, the **period** of the **sin**, **cos**, **csc**, and **sec** graphs is 2π radians, and the **period** for the **tan** and **cot** graphs is π radians. Because the circular functions are cyclical in nature, they are called **periodic functions**.

Sine Wave

A **sine wave** or **sinusoid** is a mathematical curve that describes a smooth periodic oscillation. A sine wave is a continuous wave. It is named after the function sine, of which it is the graph. It occurs often in Pure and Applied Mathematics, as well as Physics, Engineering, Signal Processing, and many other fields. Its basic form as a function of time (t) is:

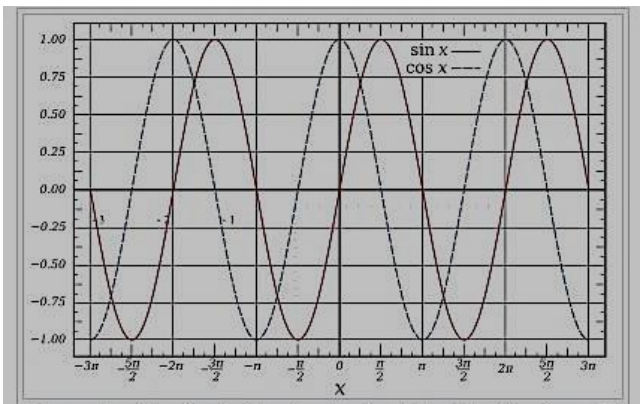
$$y(t) = A\,sin\,(2\pi\,ft + \varphi) = A\sin(\omega t + \varphi)$$

where:

- A, *amplitude*, the peak deviation of the function from zero
- f, *ordinary frequency*, the *number* of oscillations (cycles) that occur each second of time
- $\omega = 2\pi f$, *angular frequency*, the rate of change of the function argument in units of radians per second
- φ, *phase*, specifies (in radians) where in its cycle the oscillation is at $t = 0$

When φ is non-zero, the entire waveform appears to be shifted in time by the amount $\frac{\varphi}{\omega}$ seconds. A negative value represents a delay, and a positive value represents an advance.

The sine wave is important in physics because it retains its wave shape when added to another sine wave of the same frequency and arbitrary phase and magnitude. It is the only periodic waveform that has this property. This property leads to its importance in Fourier analysis and makes it acoustically unique.



The graphs of the sine (solid red) and cosine (dotted blue) functions are sinusoids of different phases

Equations of Simple Harmonic Motion

The displacement y (directed height or length) of an object behaving in a simple harmonic motion with respect to time t is given by one of the following equations:

$$y = A\ sinB(t - C) + D \text{ or } = A\ cosB(t - C) + D$$

In both equations, we have the following information:

- amplitude $=|A| = \frac{1}{2}(M - m)$ – the maximum displacement above and below the rest position or central position or equilibrium, where M is the maximum height and m is the minimum height;
- period $= \frac{2\pi}{|B|}$ – the time required to complete one cycle (from one highest or lowest point to the next);
- frequency $= \frac{|B|}{2\pi}$ – the number of cycles per unit of time;
- C – responsible for the horizontal shift in time; and
- D – responsible for the vertical shift in displacement.

Using Trigonometric Functions to Solve Real-World Scenarios

Suppose the function $y = 5tan\left(\frac{\pi}{4}t\right)$ marks the distance in the movement of a light beam from the top of a police car across a wall where t is the time in seconds and y is the distance in feet from a point on the wall directly across from the police car.

a. Find and interpret the stretching factor and period.

b. Graph on the interval [0, 5].

c. Evaluate $f(1)$ and discuss the function's value at that input.

Solution

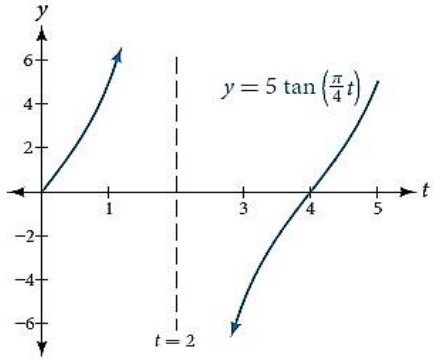
a. We know from the general form of $y = Atan(Bt)$ that $|A|$ is the stretching factor and $\frac{\pi}{B}$ is the period.

$$y = \underset{\underset{A}{\uparrow}}{5}\tan\left(\underset{\underset{B}{\uparrow}}{\frac{\pi}{4}}t\right)$$

We see that the **stretching factor is 5**. This means that the beam of light will move 5 ft after half the period.

The period is $\frac{\pi}{\frac{\pi}{4}} = \pi \cdot \frac{4}{\pi} = 4$. This means that every 4 seconds, the beam of light

sweeps the wall. The distance from the spot across from the police car grows larger as the police car approaches.



b. To graph the function, we draw an asymptote at $t = 2$ and use the stretching factor and period.

c. Period: $f(1) = 5tan\left(\frac{\pi}{4}(1)\right) = 5tan\left(\frac{\pi}{4}\right) = 5(1) = 5$; after 1 second, the beam of light has moved 5 ft from the spot across from the police car.

What's More

Solve the problem below.

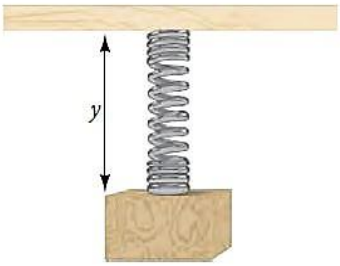
Determining a Rider's Height on a Ferris Wheel

The London Eye is a huge Ferris wheel with a diameter of 135 meters (443 feet). It completes one rotation every 30 minutes. Riders board from a platform 2 meters above the ground. Express a rider's height above ground as a function of time in minutes.

What I can Do

Solve the problem below.

A weight is attached to a spring that is then hung from a board. As the spring oscillates up and down, the position y of the weight relative to the board ranges from −1 in. (at time $x = 0$) to −7 in. (at time $x = \pi$) below the board. Assume the position of y is given as a sinusoidal function of x. Sketch a graph of the function, and then find a cosine function that gives the position y in terms of x.



Assessment

Solve the given problem.

A weight is suspended from a spring and is moving up and down in a simple harmonic motion. At start, the weight is pulled down 5 cm below the resting position, and then released. After 8 seconds, the weight reaches its highest location for the first time. Find the equation of the motion.

What I Need to Know

Expected Learning Outcomes: **(LC Codes: STEM_PC11AG-IIf-1-3)**

1. Determine whether an equation is an identity or a conditional equation;
2. Apply trigonometric identities to find other trigonometric values; and
3. Solve situational problems involving trigonometric identities.

What I know

Write the letter of the correct answer on a separate sheet of paper.

1. **Which of the following is NOT an example of identity equation?**

   A. $3x + 4 = 4 + 3x$ C. $3x (x + 1) = 3x^2 + 3$

   B. $(x - 2)^2 = x^2 - 4x + 4$ D. $2(x-1) + 3 = 3x - (x - 4)$

2. **Which of the following is NOT an identity?**

   A. $\sin^2 a + \cos^2 a = 1$ C. $1 + \cot^2 a = \csc^2 a$

   B. $\sin a = \tan a * \cos a$ D. $1 - \sec^2 a = \tan^2 a$

3. **Which of the following is an identity?**

   A. $\sin (a) \cos (a) = (1/2) \sin (2 a)$ C. $\sin(-a) = \sin a$

   B. $\sin a + \cos a = 1$ D. $\tan a = \cos a / \sin a$

4. **Evaluate: $\sin t + \cos^2 t / \sin t =$**

   A. $\sin t$ B. $\csc t$ C. $\sec t$ D. $\cos t$

5. **Which of the following is NOT an identity?**

   A. $\tan(2 t) = 2 \tan (t)$ C. $\sin(-t) = - \sin(t)$

   B. $\sin^2 a = 1 - \cos^2 a$ D. $\sec(-t) = \sec(t)$

Lesson 6 Trigonometric Identities

What's New

Identity and Conditional Equation

- *Identity* is an equation which is true for all values of the variable in the domain of the equation.

- *Conditional Equation* is an equation that is true for some values of the variables but not true for others.

Examples:

Identity	**Conditional Equation**
a. $(x^2 - 4) = (x + 2)(x - 2)$	a. $x^2 - 3x + 2 = 0$
b. $(x + 1)^2 = x^2 + 2x + 1$	b. $(x + 1)^2 = x^2 + 1$
c. $\frac{x^2-1}{x-1} = 2x - 1$	c. $\frac{x^2-1}{x-1} = x + 1$

What is It

The Trigonometric Identities

- Reciprocal Identities

$\cos\theta = \frac{1}{\sec\theta}$

$\sin\theta = \frac{1}{\csc\theta}$

$\tan\theta = \frac{1}{\cot\theta}$

- Pythagorean Identities

$\cos^2\theta + \sin^2\theta = 1$

$\sec^2\theta - \tan^2\theta = 1$

$\csc^2\theta - \cot^2\theta = 1$

- Quotient Identities

$\tan\theta = \frac{\sin\theta}{\cos\theta}$

$\cot\theta = \frac{\cos\theta}{\sin\theta}$

- Even-Odd Identities

$\cos(-\theta) = \cos\theta$

$\sin(-\theta) = -\sin\theta$

$\tan(-\theta) = -\tan\theta$

Sum and Difference Identities

$\cos(A + B) = \cos A \cos B - \sin A \sin B$

$\sin(A + B) = \sin A \cos B + \cos A \sin B$

$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A \tan B}$

$\cos(A - B) = \cos A \cos B + \sin A \sin B$

$\sin(A - B) = \sin A \cos B - \cos A \sin B$

$\tan(A - B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$

Examples: Use sum and difference identities to evaluate the following.

1. $\sin 150°$

   Solution:

   $= \sin(90° + 60°)$

   $\sin(A + B) = \sin A \cos B + \cos A \sin B$

   $\sin(90° + 60°) = \sin(90°)\cos(60°) + \cos(90°)\sin(60°)$

   $\sin(90° + 60°) = 1\left(\frac{1}{2}\right) + 0\left(\frac{\sqrt{3}}{2}\right)$

   $\boldsymbol{\sin(90° + 60°) = \frac{1}{2}}$

2. $\cos\left(-\frac{\pi}{12}\right)$

   Solution:

   $\cos\left(-\frac{\pi}{12}\right) = cos\,\frac{\pi}{12}$ $\longrightarrow$ even-odd identity

   $cos\,\frac{\pi}{12} = \cos\left(\frac{\pi}{3} - \frac{\pi}{4}\right)$ $\longrightarrow$ sum and difference identity

   $\cos\left(\frac{\pi}{3} - \frac{\pi}{4}\right) = cosA\ cosB + sinA\ sinB$

   $\cos\left(\frac{\pi}{3} - \frac{\pi}{4}\right) = cos\frac{\pi}{3}\ cos\frac{\pi}{4} + sin\frac{\pi}{3}\ sin\frac{\pi}{4}$

   $\cos\left(\frac{\pi}{3} - \frac{\pi}{4}\right) = \left(\frac{1}{2}\right)\left(\frac{\sqrt{2}}{2}\right) + \left(\frac{\sqrt{3}}{2}\right)\left(\frac{\sqrt{2}}{2}\right)$

   $\cos\left(\frac{\pi}{3} - \frac{\pi}{4}\right) = \left(\frac{\sqrt{2}}{4}\right) + \left(\frac{\sqrt{6}}{4}\right)$

   $\mathbf{\cos\left(\frac{\pi}{3} - \frac{\pi}{4}\right) = \frac{\sqrt{6} + \sqrt{4}}{4}}$

To prove a trigonometric identity:

1. Work on one side of the equation. It is usually better to start with the more complex side, as it is easier to simplify than to build.
2. Look for opportunities to factor expressions, square a binomial, or add fractions.
3. Noting which functions are in the final expression, look for opportunities to use the identities and make the proper substitutions.
4. If these steps do not yield the desired result, try converting all terms to sines and cosines.

Examples:

Prove each of the following identities:

a. $cos\theta tan\theta = sin\theta$ b. $cos\theta = sin\theta cos\theta csc\theta$ c. $cot^2\theta = cot\theta\left(\frac{1}{tan\theta}\right)$

Solution:

a. $cos\theta tan\theta = sin\theta$

$cos\theta\left(\frac{sin\theta}{cos\theta}\right) = sin\theta$

$\cancel{cos\theta}\left(\frac{sin\theta}{\cancel{cos\theta}}\right) = sin\theta$

$\boldsymbol{sin\theta = sin\theta}$

b. $cos\theta = sin\theta cos\theta csc\theta$

$cos\theta = sin\theta cos\theta\left(\frac{1}{sin\theta}\right)$

$cos\theta = \cancel{sin\theta} cos\theta\left(\frac{1}{\cancel{sin\theta}}\right)$

$\boldsymbol{cos\theta = cos\theta}$

c. $cot^2\theta = cot\theta\left(\frac{1}{tan\theta}\right)$

$cot^2\theta = cot\theta(cot\theta)$

$\boldsymbol{cot^2\theta = cot^2\theta}$

What's More

Write I if the given is an identity, and CE if conditional equation.

1. $x^2 - 2x = -1$
2. $(x^2 - 9) = (x + 3)(x - 3)$
3. $x - 5 = -5 + x$
4. $sec\theta = \frac{1}{cos\theta}$
5. $(x - 3)^2 = x^2 + 9$

What I can Do

A. Use sum and difference identities to evaluate the following.

1) $\sin\left(\frac{\pi}{2} - \frac{\pi}{3}\right)$
2) $\sec\left(\frac{\pi}{2} - \frac{5\pi}{6}\right)$
3) $\sin(90° - 120°)$
4) $\tan(90° + 240°)$
5) $\sin(90° + 60°)$

B. Use even- odd identities to evaluate the following.

1) $\csc(-240°)$
2) $sin(-\frac{11\pi}{6})$
3) $2sin(-30°)\cos(-30°)$
4) $\cos\left(-\frac{4\pi}{3}\right)$
5) $sec(-315°)$

Assessment

Prove each of the following identities.

1) $sec\theta sin\theta = tan\theta$
2) $cot\theta sec\theta = csc\theta$
3) $\frac{csc\theta}{sec\theta} = cot\theta$
4) $1 + \frac{tan\theta}{cot\theta} = sec^2\theta$
5) $\frac{1 - sin^2\theta}{sec\theta} = cos^3\theta$

What I Need to Know

Expected Learning Outcome: **(LC Codes: STEM_PC11AG-IIg-1)**

1. Illustrate the domain and range of the inverse trigonometric functions.

What I know

Write the letter of the correct answer on a separate sheet of paper.

1) **Evaluate:** $\boldsymbol{Arccos(cos\frac{4\pi}{3})}$**.**

   A. $\frac{2\pi}{3}$ B. $\frac{3\pi}{2}$ C. $\frac{\pi}{2}$ D. $\frac{\pi}{3}$

2) **Find the value of** $\boldsymbol{Arctan(-1)}$ **.**

   A. -2 B. -1 C. 1 D. 2

3) **Find the value of** $\boldsymbol{Arcsin\frac{\sqrt{3}}{2}}$ **.**

   A. $\frac{2\pi}{3}$ B. $\frac{3\pi}{2}$ C. $\frac{\pi}{2}$ D. $\frac{\pi}{3}$

4) **Evaluate:** $\boldsymbol{Arccot0}$**.**

   A. $\frac{\pi}{4}$ B. $\frac{\pi}{3}$ C. $\frac{\pi}{2}$ D. π

5) **Find the value of** $\boldsymbol{\sin(Arcsin\frac{\sqrt{3}}{2})}$

   A. $\frac{\sqrt{3}}{2}$ B. $\frac{\sqrt{2}}{2}$ C. $\frac{\sqrt{5}}{2}$ D. $\frac{\sqrt{7}}{3}$

Lesson 7 Inverse Trigonometric Functions

What's New

In your study of algebraic functions like polynomial functions, you discussed already inverse functions. It was noted that if a function is one-to- one, it will have an inverse function. However, if the function is not one- to- one, it may be possible to restrict its domain in some suitable values such as that the restricted function is one- to- one. In this section, we will use these ideas to define inverse trigonometric functions.

The Six Inverse Trigonometric Functions and how it will work.

The **Inverse sine function** with domain [-1,1] and range $[-\frac{\pi}{2},\frac{\pi}{2}]$ is defined by $y = Arcsin\ x$ if and only if $\sin y = x$.

The **Inverse cosine function** with domain [-1,1] and range $[0,\pi]$ is defined by $y = Arcscos\ x$ if and only if $\cos y = x$.

The **Inverse tangent function** with domain R and range $[-\frac{\pi}{2}, \frac{\pi}{2}]$ is defined by $y = Arctan\, x$ if and only if $\tan y = x$.

The **Inverse cotangent function** with domain R and range $[0, \pi]$ is defined by $y = Arccot\, x$ if and only if $\cot y = x$.

The **Inverse secant function** with domain $(-\infty, -1] \cup [1, +\infty)$ and range $[0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi]$ is defined by $y = Arcsec\, x$ if and only if $\sec y = x$.

The **Inverse cosecant function** with domain $(-\infty, -1] \cup [1, +\infty)$ and range $[\frac{\pi}{2}, 0) \cup (0, \frac{\pi}{2}]$ is defined by $y = Arccsc\, x$ if and only if $\csc y = x$.

What is It

Evaluate the following inverse trigonometric functions.

1) $Arcsin\ 0$

Solution:

The angle θ in the interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$ such that $\sin\theta = 0$ is 0.

Therefore, $Arcsin\ 0 = 0$.

2) $Arccos\ 0$

Solution:

The angle θ in the interval $[0, \pi]$ such that $\cos\theta = 0$ is $\frac{\pi}{2}$.

Therefore, $Arccos\ 0 = \frac{\pi}{2}$.

3) $Arctan\ 0$

Solution:

The angle θ in the interval $[-\frac{\pi}{2}, \frac{\pi}{2}]$ such that $\tan\theta = 0$ is 0.

Therefore, $Arctan\ 0 = 0$.

4) $Arccot(-1)$

Solution:

The angle θ in the interval $[0, \pi]$ such that $\cot\theta = -1$ is $\frac{3\pi}{4}$.

Therefore, $Arccot(-1) = \frac{3\pi}{4}$.

5) $Arcsec(2)$

Solution:

The angle θ in $[0, \frac{\pi}{2}) \cup (\frac{\pi}{2}, \pi]$ such that $\sec\theta = 2$ is $\frac{3\pi}{4}$.

Therefore, $Arcsec(2) = \frac{\pi}{3}$.

6) $Arccsc\sqrt{2}$

Solution:

The angle θ in $[0,\frac{\pi}{2}) \cup (\frac{\pi}{2},\pi]$ such that $\csc\theta = \sqrt{2}$ is $\frac{\pi}{4}$.

Therefore, $Arccsc\sqrt{2} = \frac{\pi}{4}$.

The six inverse trigonometric functions can be written as:

Trigonometric Functions	**Other Notation**
$y = Arcsin\ x$	$y = \sin^{-1} x$
$y = Arccos\ x$	$y = \cos^{-1} x$
$y = Arctan\ x$	$y = \tan^{-1} x$
$y = Arccot\ x$	$y = \cot^{-1} x$
$y = Arccsc\ x$	$y = \csc^{-1} x$
$y = Arcsec\ x$	$y = \sec^{-1} x$

To help you familiarize with the domain and range of the inverse trigonometric functions, we summarized them in table below.

Function	**Domain**	**Range**
$y = Arcsin\ x$	[-1,1]	$[-\frac{\pi}{2},\frac{\pi}{2}]$
$y = Arccos\ x$	[-1,1]	$[0,\pi]$
$y = Arctan\ x$	Set of real numbers	$(-\frac{\pi}{2},\frac{\pi}{2})$
$y = Arccot\ x$	Set of real numbers	$(0,\pi)$
$y = Arccsc\ x$	$(-\infty,-1] \cup [1,+\infty)$	$[0,\frac{\pi}{2}) \cup (\frac{\pi}{2},\pi]$
$y = Arcsec\ x$	$(-\infty,-1] \cup [1,+\infty)$	$[\frac{\pi}{2},0) \cup (0,\frac{\pi}{2}]$

The resulting value of an inverse trigonometric function is unique and must come only from the range of the function.

What's More

A. Evaluate the following inverse trigonometric functions.

1) $Arcsin\left(-\frac{1}{2}\right)$ 2) $Arccos\frac{\sqrt{2}}{2}$ 3) $Arctan(-1)$

4) $Arcsec\sqrt{2}$ 5) $Arccsc\sqrt{2}$ 6) $Arccot1$

B. Identify the domain and range of each equation. Leave your answer in radians.

1) $y = \sin\cot^{-1}x$ 2) $y = \tan^{-1}(\cot x)$

What I can Do

Evaluate the following inverse trigonometric functions.

1) $Arcsin\ \frac{1}{2}$ 3) $Arcsin\ \sqrt{3}$ 5) $Arcsin\ \sqrt{2}$

2) $Arcsin\ (-1)$ 4) $Arcsin\ 0$ 6) $Arcsin\ (-2)$

Assessment

Evaluate the following inverse trigonometric functions. Round off your answer in 2 decimals.

1) $Arcsin\frac{1}{2}$ 5) $Arccsc\ \frac{2\sqrt{3}}{3}$ 9) $Arctan(-8.45)$

2) $Arccos\frac{1}{2}$ 6) $Arccot\ 1$ 10) $Arcsin\ (0.2447)$

3) $Arctan\frac{\sqrt{3}}{3}$ 7) $Arcsin\ 0$

4) $Arcsec\ 2$ 8) $Arccos\ 0$

What I Need to Know

Expected Learning Outcome: **(LC Codes: STEM_PC11AG-IIa-1-2)**

1. Solve trigonometric equations.

What I know

Solve the given equations.

1. **sin x = $\frac{\sqrt{3}}{2}$ in the interval [0,2π)**

   A. $\frac{\pi}{3}, \frac{5\pi}{6}$ B. $\frac{\pi}{4}, \frac{2\pi}{3}$ C. $\frac{\pi}{4}, \frac{5\pi}{6}$ D. $\frac{\pi}{3}, \frac{2\pi}{3}$

2. **$\sin^2 x - 1 = 0$ in the interval [0,2π)**

   A. $\frac{\pi}{3}, \frac{5\pi}{3}$ B. $\frac{\pi}{2}, \frac{3\pi}{2}$ C. $\pi, \frac{7\pi}{4}$ D. $\frac{\pi}{4}, \frac{5\pi}{6}$

3. **sec x = 2 in the interval [0,2π)**

   A. $\frac{\pi}{3}, \frac{5\pi}{3}$ B. $\frac{\pi}{2}, \frac{3\pi}{2}$ C. $\pi, \frac{7\pi}{4}$ D. $\frac{\pi}{4}, \frac{5\pi}{6}$

4. **2cos x = 1 in the interval [0,2π)**

   A. $\frac{\pi}{3}, \frac{5\pi}{3}$ B. $\frac{\pi}{2}, \frac{3\pi}{2}$ C. $\pi, \frac{7\pi}{4}$ D. $\frac{\pi}{4}, \frac{5\pi}{6}$

5. **Cos x sin x + cos x = 0 in the interval [0,2π). Find the value of x.**

Lesson 8 — Trigonometric Equations

What's New

Equations involving trigonometric functions of a variable are known as trigonometric equations. The solutions to a trigonometric equation are values of the angles which satisfy the equation.

Recall that a solution or root of an equation is a number in the domain of the equation that, when substituted to the variable, makes the equation true. The set of all the solutions of an equation is called the solution set of the equation.

Solving Trigonometric Equations:

If the equation contains only one trigonometric term, isolate that term, and solve for the variable.

Example 1: Solve 2 sec x - 4 = 0 in the interval $[0, 2\pi)$.

Solution:	$2\sec x - 4 = 0$	Write the original equation.
	$2\sec x - 4 + 4 = 0 + 4$	Add 4 to each side.
	$(\frac{1}{2})\ 2 \sec x = 4\ (\frac{1}{2})$	Using MPE, multiply each side by $\frac{1}{2}$.
	$\sec x = 2$	Simplify.
	$\frac{1}{\sec x} = \frac{1}{2}$	Write its reciprocal.
	$\cos x = \frac{1}{2}$	Get the values in the unit circle.

Answer: x = $\frac{\pi}{3}, \frac{5\pi}{3}$

If the equation is quadratic in form, rewrite the equation to have zero on one side then, use factoring, finding square roots, or the quadratic formula.

Example 2: Solve: $2\sin^2 x - \sin x - 1 = 0$ in the interval $[0, 2\pi)$

Solution:	$2\sin^2 x - \sin x - 1 = 0$	Write the original equation.
	$(2 \sin x + 1)(\sin x - 1) = 0$	Factor.
	$(2 \sin x + 1) = 0$ and $(\sin x - 1) = 0$	Apply the Zero Product Property.

$2 \sin x + 1 = 0$ or $\sin x - 1 = 0$

$2 \sin x = -1$ $\qquad$ $\sin x = 1$

$(\frac{1}{2})\ 2 \sin x = -1\ (\frac{1}{2})$

$\sin x = -\frac{1}{2}$

Answer: x = $\frac{7\pi}{6}, \frac{11\pi}{6}, \frac{\pi}{2}$

If the equation contains more than one trigonometric function, try to express everything in one term by manipulation and the use of identities.

What is It

Solving Trigonometric Equations with Two or More Terms

In cases where the equation has multi – step techniques of finding their solutions, we may express the equation in terms of **only one** circular function by manipulation, and use identities that we have learned in the previous lesson.

Example 3: Solve for $x \in [0, 2\pi)$: $3 \sin^2 x + \cos 2x - 2 = 0$

Solution:	$3 \sin 2 x + \cos 2x - 2 = 0$	Write the original equation.
	$3 \sin^2 x + 1 - 2 \sin^2 x - 2 = 0$	Use the identity $\cos 2x = 1 - 2 \sin^2 x$.
	$\sin^2 x - 1 = 0$	Combine like terms.

$\sin^2 x - 1 + 1 = 0 + 1$	Using APE, add each side by 1.
$\sin^2 x = 1$	Simplify.
$\sin x = \pm 1$	Take the square root.

Answer: x = $\frac{\pi}{2}, \frac{3\pi}{2}$

Example 4: Solve: $\cos^2 x + \cos x - 2 = 0$ in the interval $0 \leq x \leq 2\pi$

Solution:

$\cos^2 x + \cos x - 2 = 0$	Write the original equation.
$(\cos x + 2)(\cos x - 1) = 0$	Use factoring technique.
$(\cos x + 2) = 0 \quad (\cos x - 1) = 0$	Apply Zero Product Property.
$\cos x + 2 = 0$ or $\cos x - 1 = 0$	
$\cos x = -2 \quad \cos x = 1$	

Answer: x = 0, 2π

cos x = - 2 has no solution since -2 is not in the interval.

What's More

Find the solutions in the interval $[0, 2\pi)$.

1) **$2 \sin x + 1 = 0$**

Solution:

_____	Write the original equation.
_____	Using APE, add each side by – 1.
_____	Simplify.
_____	Using MPE, multiply each side by $\frac{1}{2}$.
_____	Simplify.

Answer: _____

2) **$2 \cos^2 x - 3 \cos x - 2 = 0$**

Solution:

_____	Write the original equation.
_____	Use factoring technique.
_____	Apply Zero Product Property.
_____	
_____	
_____	

Answer: _____

_____ has no solution since __ is not in the interval.

What I can Do

Solve the following trigonometric equations in the interval [0,2π).

1) $\sin x - 1 = 0$
2) $\sin x = \frac{1}{2}$
3) $2 \cos x - \sqrt{3} = 0$
4) $2 \sin^2 x - 3 \sin x + 1 = 0$

Assessment

Choose the letter of the correct answer.

1. **Solve for [0,2π): 2 sin x + 3 = 2**

   A. $\frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$ B. $\frac{7\pi}{6}, \frac{11\pi}{6}$ C. $\frac{4\pi}{3}, \frac{5\pi}{3}$ D. $\frac{3\pi}{4}, \frac{7\pi}{4}$

2. **Solve for [0,2π): 4 cos² x = 1**

   A. $\frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$ B. $\frac{7\pi}{6}, \frac{11\pi}{6}$ C. $\frac{4\pi}{3}, \frac{5\pi}{3}$ D. $\frac{3\pi}{4}, \frac{7\pi}{4}$

3. **Solve for [0,2π): tan x + 1 = 0**

   A. $\frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$ B. $\frac{7\pi}{6}, \frac{11\pi}{6}$ C. $\frac{\pi}{2}, \pi, \frac{3\pi}{2}$ D. $\frac{3\pi}{4}, \frac{7\pi}{4}$

4. **Solve for [0,2π): cos² x + cos x = 0**

   A. $\frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$ B. $\frac{7\pi}{6}, \frac{11\pi}{6}$ C. $\frac{4\pi}{3}, \frac{5\pi}{3}$ D. $\frac{\pi}{2}, \pi, \frac{3\pi}{2}$

5. **Solve for [0,2π): 2 sin x + $\sqrt{3}$ = 0**

   A. $\frac{3\pi}{4}, \frac{7\pi}{4}$ B. $\frac{\pi}{2}, \pi, \frac{3\pi}{2}$ C. $\frac{4\pi}{3}, \frac{5\pi}{3}$ D. $\frac{\pi}{2}, \pi, \frac{3\pi}{2}$

What I Need to Know

Expected Learning Outcome: **(LC Codes: STEM_PC11T-IIi-2)**

Solve situational problems involving inverse trigonometric functions and trigonometric equations.

What I know

Write the letter of the answer on a separate sheet.

1) A man with a height of 5 feet casts a shadow on the ground. If the shadow is 2 feet long, find the angle of elevation of the sun from the tip of the shadow.
   A. 69. 20° B. 68.20° C. 67.20° D. 65.20°

2) A cargo ship is being observed 100 ft above the sea level from the top of a watch tower. If the angle of depression of the cargo ship from the top of the watch tower is 30°, how far is the cargo ship from the watch tower?
   A. 200 B. 300 C. 400 D. 500
3) Two buildings were being observed from an aircraft. The angles of depression of building ALPHA and building BETA are 40° and 55°, respectively. If building ALPHA is 3km away from building BETA, find the distance of the aircraft from building ALPHA.
   A. 95° B. 85° C. 75° D. 65°
4) A safety regulation stated that the maximum angle of elevation for a rescue ladder is 72°. A fire department's longest ladder is 110 feet. What is the maximum safe rescue height?
   A. 101.6 ft B. 102.6 ft C. 103.6 ft D. 104.6 ft
5) Two cargo vessels are being observed from a small island. Ship Delta is located 70 nm (nautical miles) N25° E from island while Ship Lambda is located 85 nm S63° from the island. What is the distance between Ship Delta and Ship Delta?
   A. 101.6 nm B. 111.98 nm C. 103.6 nm D. 104.6 nm

Lesson 8 — Solving Problems involving Trigonometric Functions

What's New

Many situations in our daily life involve mathematical concepts. Engineering and scientific theories can be verified through calculations. Distance or length which is inaccessible to measure can be determine using trigonometric functions.

We can apply the trigonometric ratios to solve real life problems involving triangles. It is also important to note that the sum measures of the angles of a triangle is equal to 180°. Law of sines and cosines are also of great help to solve problems in navigation. Usually, directions in navigation are expressed in terms of bearing.

What is It

The Six Trigonometric Ratios:

Let θ be an acute angle of a right triangle.

$$sin\theta = \frac{side\ opposite\ \theta}{hypotenuse} \qquad cos\theta = \frac{side\ adjacent\theta}{hypotenuse} \qquad tan\theta = \frac{side\ opposite\ \theta}{side\ adjacent\theta}$$

$$csc\theta = \frac{hypotenuse}{side\ opposite\ \theta} \qquad sec\theta = \frac{hypotenuse}{side\ adjacent\theta} \qquad cot\theta = \frac{side\ adjacent\theta}{side\ opposite\ \theta}$$

Example:

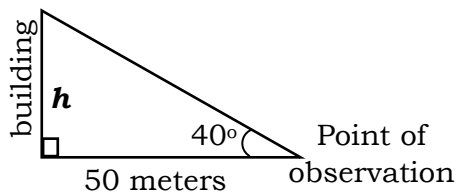
From a point on the ground 50 meters away from the base of a building, the angle of elevation of the roof top of the building is 40°. How high is the building?

Solution:

side opposite of θ is h, $\theta = 40^o$
side adjacent to θ is 50 meters

Solution:

Illustration:



Let h be the height of the building
To use the given data, we use tangent function.

$$tan40° = \frac{h}{50\ meters}$$

$$h = (50\ meters)\ (tan40°)$$

$$h = 41.95\ meters$$ The building is 41.95 meters.

What's More

Let's try this,

Solve the following problems. Show your solutions on a separate sheet.

1) In a right triangle, the 58-cm hypotenuse makes a 51-degree angle with one of the legs. To the nearest tenth of a cm, how long is that leg? **Ans. 36.5 cm**
2) A 14-foot ladder is used to scale a 13-foot wall. At what angle of elevation must the ladder be situated to reach the top of the wall? **Ans. θ =68.2 °**

What I can Do

Directions: In a separate sheet of paper, solve the following problems.

1) From the top of a vertical cliff 40 m high, the angle of depression of an object that is level with the base of the cliff is 34°. How far is the object from the base of the cliff?
2) A man flies a kite and lets out 100 feet of string. The angle of elevation of the string is 52°. How high off the ground is the kite? How far away is the man from the spot directly under the kite?
3) The diagonal of a rectangle is 15 cm, and the perimeter is 38 cm. What is the area?
4) One of the legs of a right triangle is twice as long as the other, and the perimeter of the triangle is 28. Find the lengths of all three sides, to three decimal places.
5) Brian's kite is flying above a field at the end of 65 m of string. If the angle of elevation to the kite measures 70°, how high is the kite above Brian's head?

Assessment

Directions: Solve the following problems. Show your solutions on a separate sheet.

1. A woman is standing on the ground at a point 78ft from the base of a building. The angle of elevation to the top of the building is 57°. To the nearest foot, how high is the building?
2. An observer on a cliff 1000 dm above sea level sights two ships due east. The angles of depression of the ships are 47˚and 32˚. Find, to the nearest decimeter, the distance between the two ships.
3. The sun shines on a flagpole, causing a shadow to be cast on the ground. The distance from the base of the pole to the tip of the shadow is 49 feet. At that time of day, the sun's rays make an angle of 38° with the ground. How tall is the flagpole?
4. A 200 ft high television transmitting tower is to be supported by guy wires running from the ground to the top of the tower. The wires make an angle of 63° with the ground.
   a) How long is each wire?

b) How far from the base of the tower must they meet the ground?

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